

The Self-Righting Automatic Dumpster Cleaner

Patent Pending: Application No. 63/723,626 (Confirmation No. 7443)

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Executive Summary: *The Self-Righting Automatic Dumpster Cleaner is a patent-pending, eco-friendly, and disposable cleaning solution designed to simplify dumpster and outdoor rolling trash bin sanitation. Its innovative features include advanced foaming technology, self-righting stability, and a user-friendly pull-tab activation system, making it a highly efficient and environmentally responsible product.*

Mission Statement: *To revolutionize waste management by offering a convenient, sustainable, and affordable cleaning solution for households, businesses, and municipalities.*

Business Model

Product Lines

1. Residential Model:

- Designed for outdoor rolling trash bins (30–50 gallons).
- Sold in packs of **5**, **10**, or **20** units.

2. Commercial Model:

- Designed for industrial dumpsters (2–10 cubic yards).
- Sold in packs of **3** or **5** units.

Revenue Streams

1. **Retail Sales:** Placement in major retailers like Walmart, Home Depot, and Target alongside trash bags.
2. **Online Sales:** Available through Amazon and the company's direct-to-consumer platform.
3. **Wholesale Distribution:** Targeting municipalities, property managers, and commercial enterprises.

Pricing

Residential Packs:

- **5-pack:** \$15
- **10-pack:** \$25
- **20-pack:** \$40

Commercial Packs:

- **3-pack:** \$75
- **5-pack:** \$120

Incentive: Bulk pricing offers minor discounts to encourage larger purchases.

Product Description

How It Works

1. Activation Mechanism:

- **Pull-Tab System:** A durable tab mechanically activates the internal cleaning solution when pulled.
- **Impact Trigger:** Activates the cleaning process upon contact with the dumpster floor.
- **Backup Timer:** Ensures activation within **5 seconds** of deployment, even if impact doesn't occur.

2. Self-Righting Technology:

- Weighted base ensures that the device remains upright during activation for maximum spray coverage.

3. Foaming Technology:

- Combines a **citric acid**, **SLS**, and **hydrogen peroxide** reaction to create a high-expansion foam.
- Foam clings to walls, floors, and lids, dissolving grime and neutralizing odors.

4. Eco-Friendly Composition:

- Biodegradable materials align with environmental sustainability goals, decomposing within **6–12 months**.

Key Specifications

Residential Model:

- **Dimensions:**
 - Diameter: **6 inches**
 - Height: **8 inches**
 - Weight: **0.6 lbs**
- **Cleaning Agent Volume:** **250 mL**

Commercial Model:

- **Dimensions:**
 - Diameter: **12 inches**
 - Height: **15 inches**
 - Weight: **2.5 lbs**
- **Cleaning Agent Volume: 1 L**

Foaming Technology

Core Components

1. **Foaming Agent:**
 - **Sodium Lauryl Sulfate (SLS):** Produces high-expansion foam.
 - Stabilized with xanthan gum for surface adhesion.
2. **Cleaning Agents:**
 - **Citric Acid:** Breaks down grease and grime.
 - **Sodium Bicarbonate:** Neutralizes odors and enhances reaction.
3. **Catalyst:**
 - **Encapsulated Hydrogen Peroxide:** Reacts with citric acid to release oxygen bubbles, expanding foam volume.

How It Works

1. **Activation:**
 - Pull-tab removes the internal barrier between cleaning agents and catalyst.
 - A spring-loaded mechanism punctures the hydrogen peroxide chamber, starting the reaction.
2. **Foam Generation:**
 - Oxygen bubbles stabilize within the surfactant-rich solution, creating dense foam.
 - Foam expands **20x–30x**, adhering to vertical surfaces for **5–7 minutes**.
3. **Cleaning:**
 - Citric acid and sodium bicarbonate dissolve organic residues, grease, and grime.
 - Deodorizing agents neutralize odors, leaving a fresh scent.

Market Analysis

Target Customers

1. **Residential:**
 - Homeowners and renters with outdoor rolling trash bins.
2. **Commercial:**
 - Restaurants, retail chains, and warehouses requiring regular dumpster sanitation.
3. **Municipalities:**
 - City sanitation departments seeking scalable, eco-friendly cleaning solutions.

Market Size

- **Residential:** 125 million trash bins in the U.S.
- **Commercial:** 2 million dumpsters in the U.S.

Competitive Edge

1. **Innovation:**
 - Self-righting design ensures reliability and effectiveness.
 - Patented foaming technology offers superior cleaning compared to traditional methods.
2. **Sustainability:**
 - Fully biodegradable, addressing growing environmental concerns.
3. **Cost Efficiency:**
 - Single-use design eliminates the need for costly professional cleaning services.

Operations Plan

Manufacturing

1. **Components:**
 - **Outer Casing:** Injection-molded biodegradable HDPE or PLA plastic.
 - **Foam Chamber:** Sealed compartments for cleaning agents and catalyst.
 - **Activation System:** Includes pull-tab and spring-loaded trigger.
2. **Assembly Process:**
 - Mold biodegradable outer casing.
 - Assemble aerosol or foam chambers.

- Install weighted base for self-righting stability.
- Attach activation system (pull-tab and impact trigger).
- Conduct quality control testing.

Logistics

- **Distribution:** Nationwide shipping via third-party logistics providers.
- **Warehousing:** Eco-friendly packaging reduces storage requirements and waste.

Financial Projections

Revenue

1. **Residential Sales:** \$1.2M
 2. **Commercial Sales:** \$850,000
- Total Revenue: \$2.05M**

Expenses

1. Manufacturing and Packaging: **\$200,000**
 2. Marketing: **\$75,000**
 3. Logistics: **\$60,000**
- Total Expenses: \$335,000**

Profit Margin:

68% Gross Profit Margin

Funding Requirements

Seeking an investment of **\$300,000** for:

1. Finalizing product design and testing.
2. Scaling up manufacturing operations.
3. Launching marketing campaigns.

Milestones

1. **Prototype Development** (Months 1–3): Validate product performance through field testing.
2. **Manufacturing Setup** (Months 4–6): Establish production lines and finalize supplier contracts.
3. **Retail Launch** (Months 6–12): Distribute products to retail stores and online platforms.

Conclusion

The **Self-Righting Automatic Dumpster Cleaner** is a groundbreaking innovation poised to redefine waste management. With its patent-pending technology, eco-friendly design, and clear market demand, it offers an unparalleled opportunity for scalability and profitability.

